7th Grade Math
2026-2027

425/35 AE

Name Jordan
Section 7- 3

Unit 2 Quiz- Probability and Fractions



Directions:

- Show all of your work on this paper. If you use another sheet of paper, please put your name on it and staple it to the quiz.
- Circle your answers.
- You can use a calculator on this quiz, but you must write down what you entered into your calculator.
- Round any decimal answers to the nearest hundredth.
- Write all fraction answers in simplest form.

For #1-4, perform the following operations with fractions. Your final answer should be a fully simplified fraction or mixed number, not an improper fraction.

(1.) $\frac{1}{6} \cdot \frac{3}{4}$ $= \frac{3}{24} = \frac{1}{8}$ ✓

(2.) $2\frac{2}{3} - \frac{3}{5}$ $= \frac{31}{15} = 2\frac{1}{15}$ ✓

$\frac{40}{15} - \frac{9}{15} = \frac{31}{15}$

(3.) $\frac{9}{2} + \frac{7}{9}$ $= \frac{95}{18} = 5\frac{5}{18}$ ✓

$\frac{81}{18} + \frac{14}{18} = \frac{95}{18}$

(4.) $\frac{2}{5} \div \frac{10}{12}$ $= \frac{12}{25}$ ✓

$\frac{2}{5} \times \frac{12}{10} = \frac{24}{50} = \frac{12}{25}$

+12/12

Roulette is a game where players bet on which number (and color) a ball will land on when the wheel (shown to the right) is spun. The wheel has 38 equal sections. For #5-6, calculate the following probabilities based on a single spin of the wheel. Your answers should be fully simplified fractions.



(5.) P(Black) = $\frac{18}{38} = 47.37\%$

(-1) reduce

(6.) P(number higher than 30) = $\frac{7}{38} = 18.42\%$

(-1)

(7.) There are different colors of marbles in a bag. Without looking, a player pulls a marble from the bag. Then, they put the marble back into the bag. Then, they pull out another marble. To win, they must not pull a purple marble either time. Is this an example of dependent or independent probability? **Explain your answer in words.**

It's dependent because if you pull a purple on the first turn you can't do the second turn. +

+ -2

(8.) Gene and Jaques like to play scrabble together. Gene has won five of the last six games they have played. Is 5/6 the experimental or theoretical probability that Gene will win the next game they play?

Theoretical + (-1)

because he not guaranteed that Gene will win

+5
10

(9.) 40% of Live Oak Students live within 1 mile of school. 4/7 of students live between 1 and 5 miles from school. What percent of students live more than 5 miles from school?

2.86 %

$\frac{4}{7}$ = 57.14%

97.14 + 2.86 = 100%

At the restaurant Big Burgers, customers have a choice of patty (beef or vegetarian), a choice of bread (white or wheat), and a choice of side dish (fries, onion rings, or fruit). Use this information to answer questions #10-11.

(10.) How many different meals could you order? Draw a tree diagram to show this.



20 different orders

-1

(11.) Anabel only wants a vegetarian burger (not beef) and she doesn't want fries. How many different options does she have?

She now has 7 options

x -1

+5

7

(12.) You can win Esther's game by rolling a "3" on a standard six-sided die, flipping "heads" on a coin, and pulling a red card (hearts or diamonds) from a standard deck of playing cards. (A standard deck of playing cards has 52 cards.) What is the theoretical probability of winning Esther's game?

$\frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{18} = \frac{1}{216}$

$\frac{1}{216} = 4.63\%$

(13.) At the 7th Grade Probability Carnival, Thalia wants to play both Oliver and Olivia's games. Oliver has calculated that the odds of winning his game is 32% while Olivia has calculated that the odds of winning her game are 18%. What are the chances that Thalia wins either (one or both) of their games? **Explain in words** how you found your answer.

$\frac{32}{100} \cdot \frac{18}{100} = \frac{5}{100}$

Because

Because if ya add them up it's 50

+3
6

★★★ *End of Quiz* ★★★